

Should I Work Out Today?

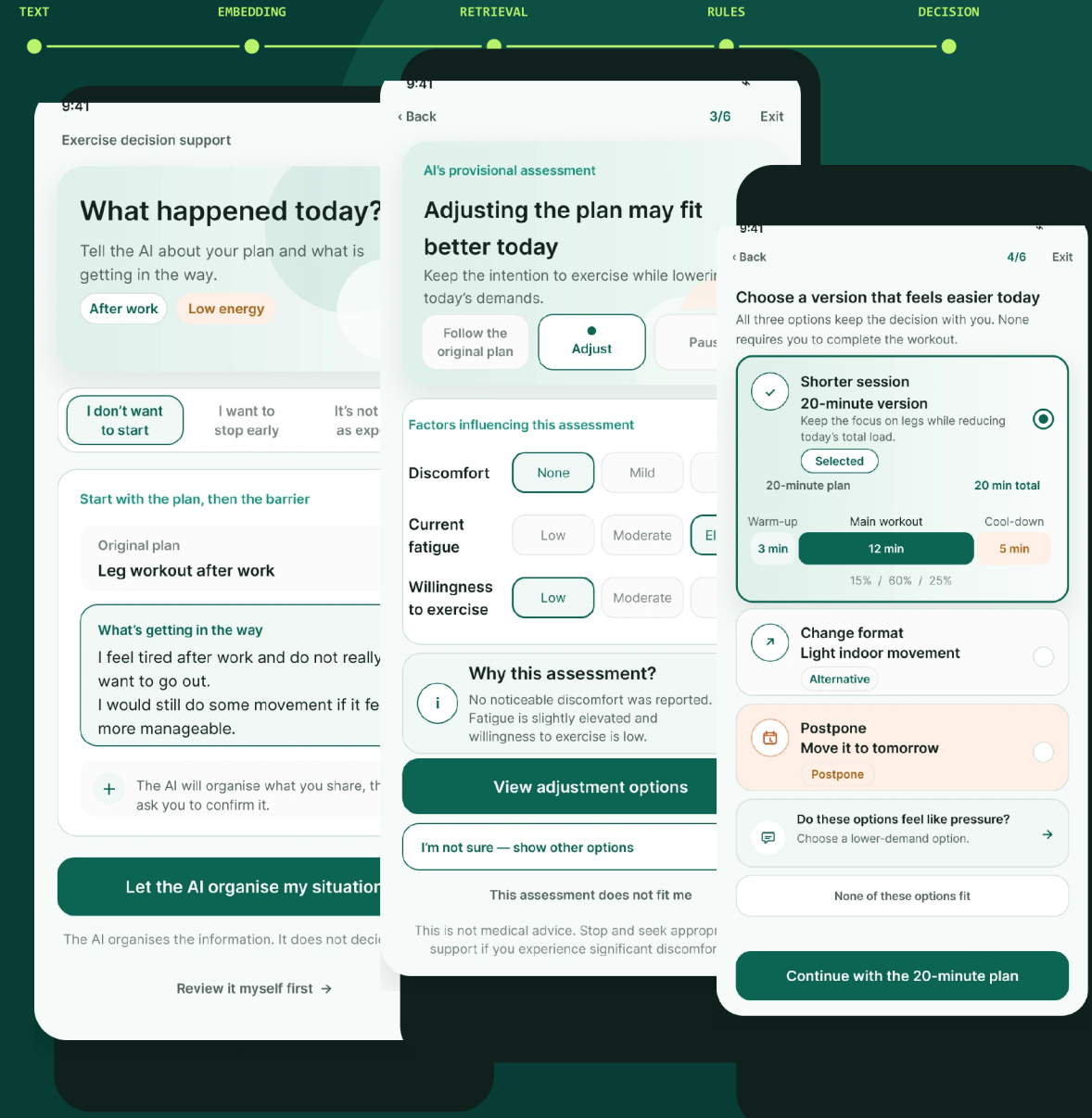
A transparent exercise decision-support experiment

- HUMAN-CENTRED AI
- ML TECHNICAL FEASIBILITY
- PRODUCT / UX

One final project, evidenced through two separate strands: a high-fidelity Figma prototype and an inspectable Python ML experiment.

SYNTHETIC FEASIBILITY EVIDENCE

Not deployed. Not clinical. Not connected to the Figma prototype.



One decision moment, investigated through two forms of evidence

PROJECT PREMISE

When a plan made earlier no longer matches time, energy, motivation or physical state, the task is not adherence. It is deciding what remains appropriate now.

RESEARCH QUESTION

How can AI-assisted decision support help users adapt a planned workout when their current state no longer matches the original plan, while preserving transparency, autonomy and safety?

HUMAN INTERFACE EVIDENCE

High-fidelity Figma prototype

Seven-screen Final English V1.5 decision flow with visible interpretation, correction routes, options, privacy choice and final user authority.

FORMATIVE USABILITY EVIDENCE

TECHNICAL FEASIBILITY EVIDENCE

Separate inspectable Python experiment

42 synthetic reference scenarios, MiniLM and TF-IDF retrieval, intent rules, safety gate, tests and frozen evaluation outputs.

NOT LIVE-CONNECTED TO FIGMA

STATUS: prototype + separate ML experiment | not deployed | not clinical | not a complete application

The plan was made earlier. The decision happens now.

08:00 / MORNING PLAN

Leg day after work

The intention is clear, but it is based on a future version of the user.

18:30 / CURRENT REALITY

The state has changed

Tired, reluctant to go outside and unsure whether changing the plan means giving up.

THE DECISION GAP

What fits now?

Continue, adapt, postpone, decline, or stop because of a safety concern.



DESIGN POSITION

The unmet need may not be another reminder.

It may be support in adapting a plan to the user's current state.

ADAPT

POSTPONE

DECLINE

SAFETY STOP

Four research lenses constrain the design

EXERCISE AUTONOMY

Avoid treating adherence as the only success.

Teixeira et al. (2012)
Ntoumanis & Moller (2025)

CONTEXT-SENSITIVE SUPPORT

State changes create decision points, tailoring variables and rules.

Nahum-Shani et al. (2018)
Hardeman et al. (2019)

HUMAN-AI INTERACTION

Interpretation, explanation, correction and control must be designed together.

Amershi et al. (2019)
Miller (2019)
Bucinca et al. (2021)

SAFETY / GOVERNANCE

Risk is contextual and cannot be reduced to one accuracy figure.

WHO (2021)
NIST AI RMF (2023)

LITERATURE INFORMED THE DESIGN POSITION. IT DID NOT SUPPLY USER OUTCOMES THE PROJECT DID NOT COLLECT.

The project narrowed until one decision became researchable



FINAL SCOPE

Support a single uncertain choice without moral ranking.

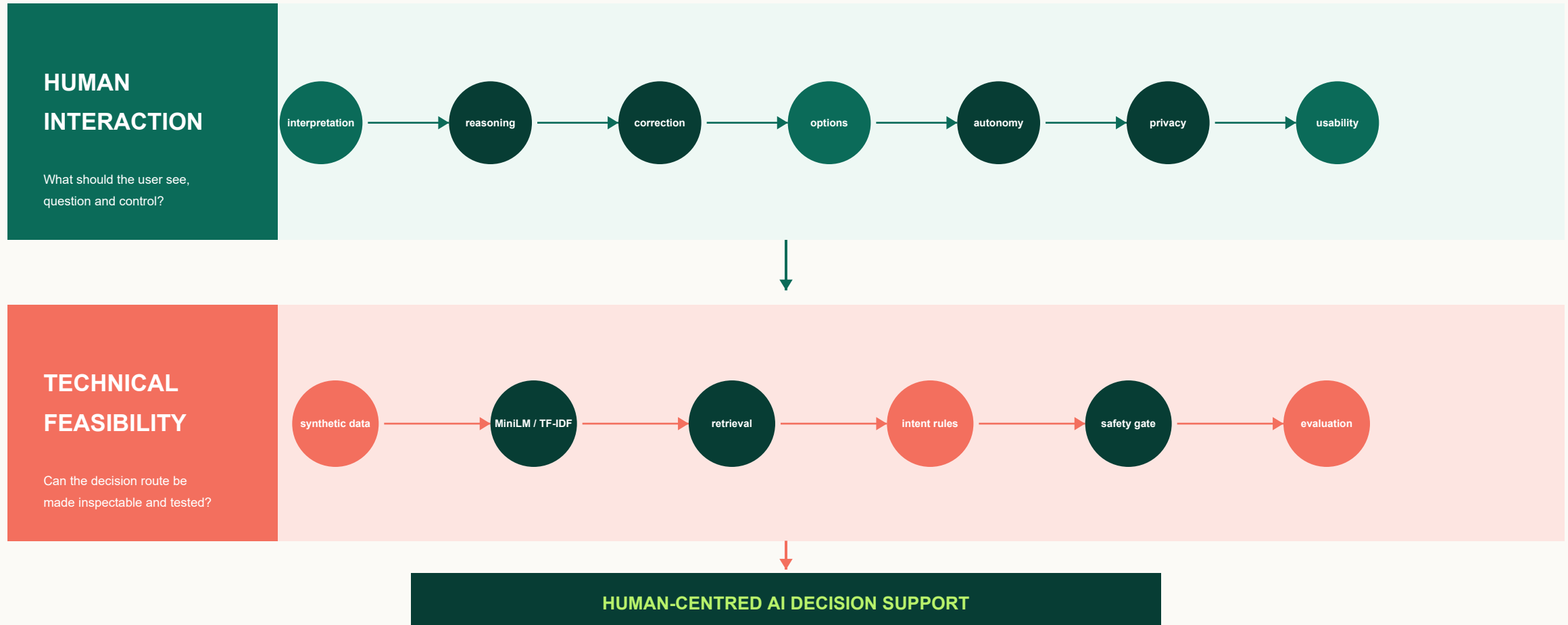
Not a workout-plan generator. Not performance tracking. Not an adherence-maximisation app.

Six constraints shape one decision



These are design constraints. They are not evidence that every interaction was validated.

System logic and interface logic answer different questions



THE ML ENGINE WAS NOT CONNECTED LIVE TO FIGMA.

Repository evidence becomes a working system

REAL ml/ ARTEFACTS

REPOSITORY / ml/

```
ml/
|-- src/
|   |-- decision_engine.py
|   |-- build_embeddings.py
|   |-- evaluate.py
|   |-- robustness_evaluate.py
|   |-- final_holdout_evaluate.py
|-- tests/ [4 files / 40 methods]
|-- data/
|   |-- exercise_scenarios.csv [42]
|   |-- evaluation_cases.csv [16]
|   |-- robustness_cases.jsonl [48]
|   |-- final_holdout_cases.jsonl [60]
|-- outputs/
|   |-- evaluation_results.json
|   |-- robustness_results.json
|   |-- final_holdout_results.json
|   |-- sample_decision.json
|   |-- sample_safety_override.json
|-- README.md
```

01 DATA

CSV / JSONL / JSON
synthetic scenarios

CSV A01 | adapt | safety=false

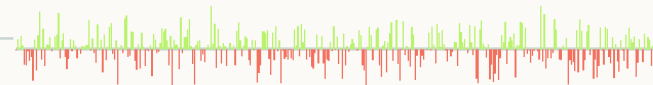
I planned a full workout after work, but I feel mentally tired and the session now feels too demanding.

```
JSONL "case_id":"H-EN-S07"
      "expected_category":"safety_stop"
```

02 REPRESENTATION

Python / Sentence Transformers
MiniLM-L6-v2 / TF-IDF

MINILM / ACTUAL A01 384-D VECTOR STRIP



TF-IDF / REAL FEATURE FORMS

w:planned	w:workout
b:planned_full	b:after_work
c3:pla	c4:work

03 DECISION

cosine / rank weighting
intent rules / safety gating

top-3 retrieval

intent

safety

final

04 VALIDATION

automated tests / SHA-256
Git / Figma / Codex-assisted

test_safety_gated_engine.py

FROZEN_*_SHA256

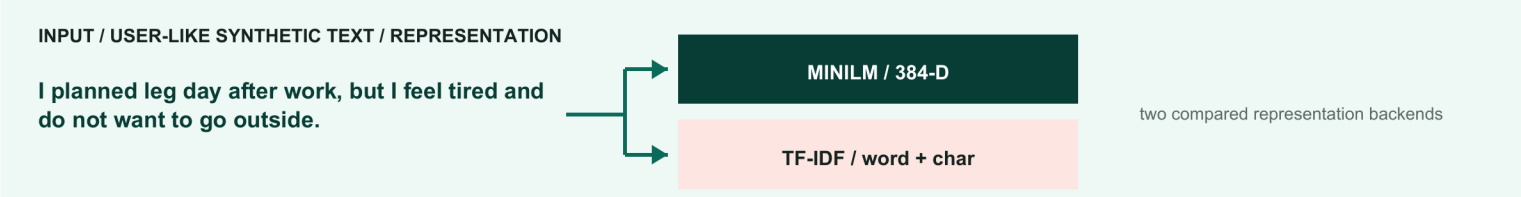
final_holdout_results.json

CODE + DATA + RESULTS + INTERFACE

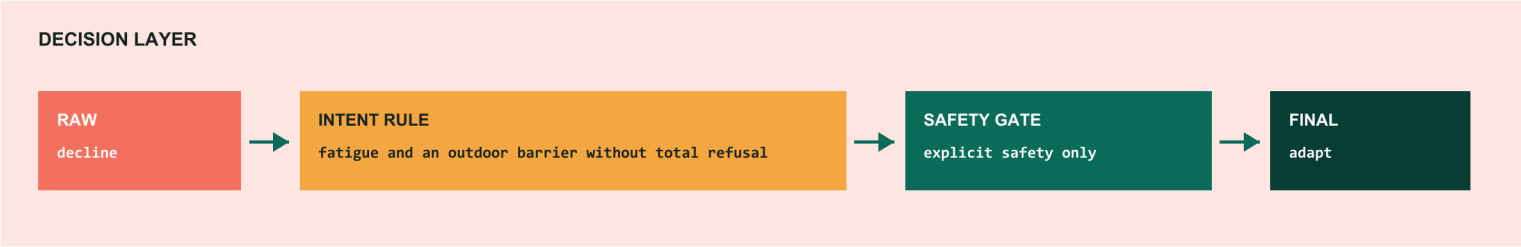
Python · Sentence Transformers · MiniLM-L6-v2 · TF-IDF · Git · Figma · Codex-assisted implementation

The pipeline is visible at every layer

REAL CODE + CHECKED-IN TRACE



RETRIEVAL / TOP-3 COSINE NEIGHBOURS				
#1	D01	decline		0.6173
#2	A01	adapt		0.5885
#3	D02	decline		0.5715



SIMILARITY IS TEXTUAL CLOSENESS, NOT CALIBRATED CONFIDENCE

RETRIEVAL

embedding_prediction_raw

top1_similarity

top2_similarity

similarity_margin

DECISION

final_prediction

decision_layer

matched_intent_rule

OVERRIDE / UNCERTAINTY

override_reason

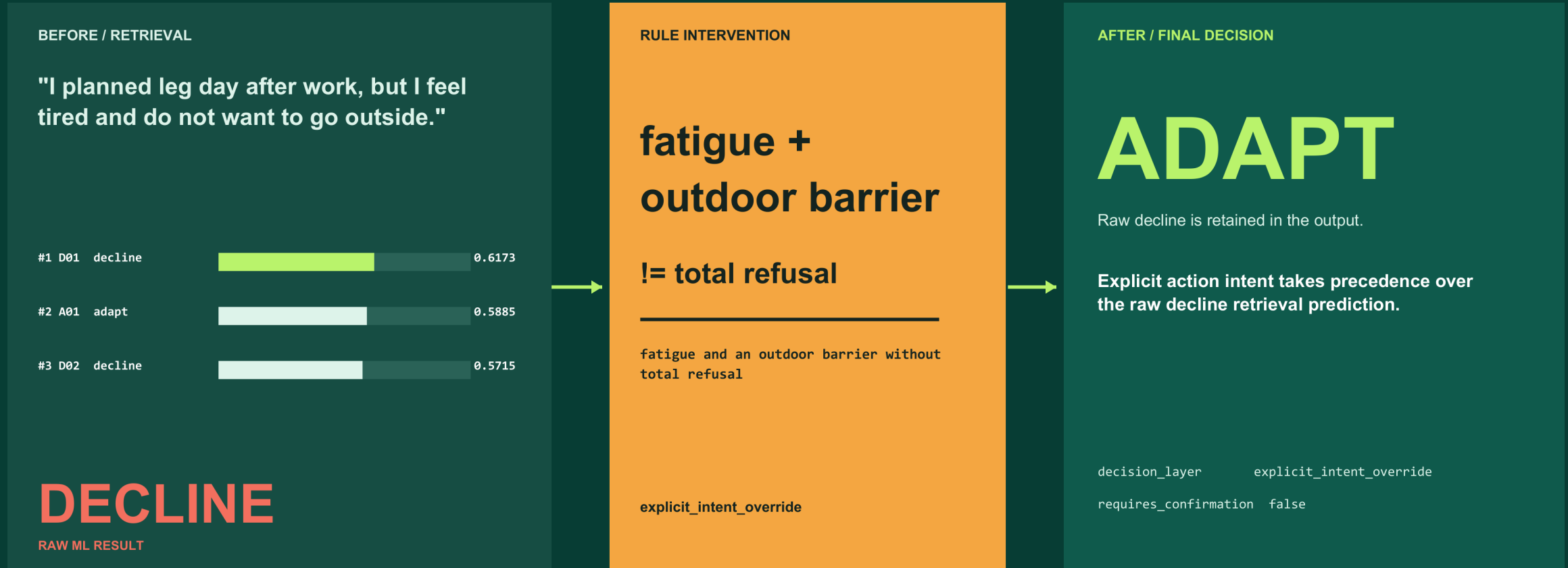
requires_confirmation

decision_engine.py / SAFETY GATE EXCERPT

```
625     if raw_safety_without_rule:
626         final_prediction = embedding_result["best_non_safety_prediction"]
627         response = CATEGORY_RESPONSES[final_prediction]
628         return {
629             **embedding_result,
630             "decision_layer": "safety_gate_override",
631             "final_prediction": final_prediction,
632             "override_reason": (
633                 "Retrieval suggested a safety category, but no explicit "
634                 "safety signal was detected."
```

An inspectable rule changes the raw result

SOURCE: sample_decision.json



RAW DECLINE -> INSPECTABLE INTENT OVERRIDE -> FINAL ADAPT

The failure changes who can authorise safety_stop



16

DEVELOPMENT

48

ROBUSTNESS

GATE

REVISION

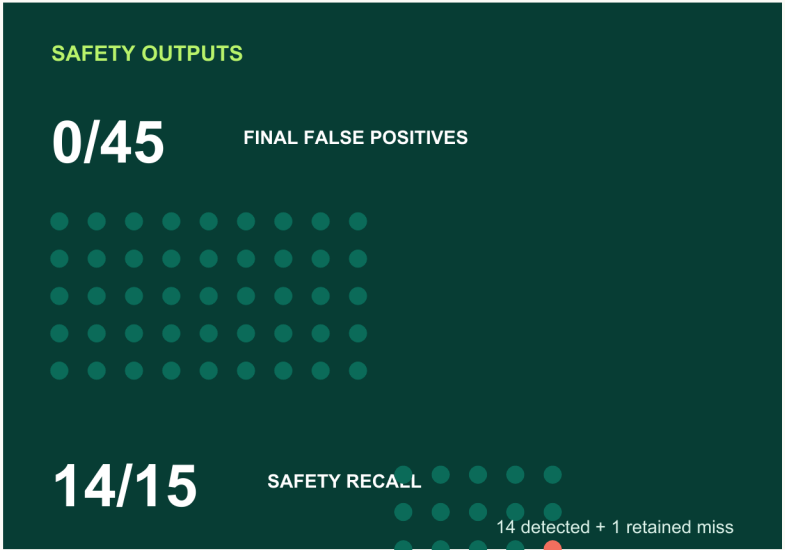
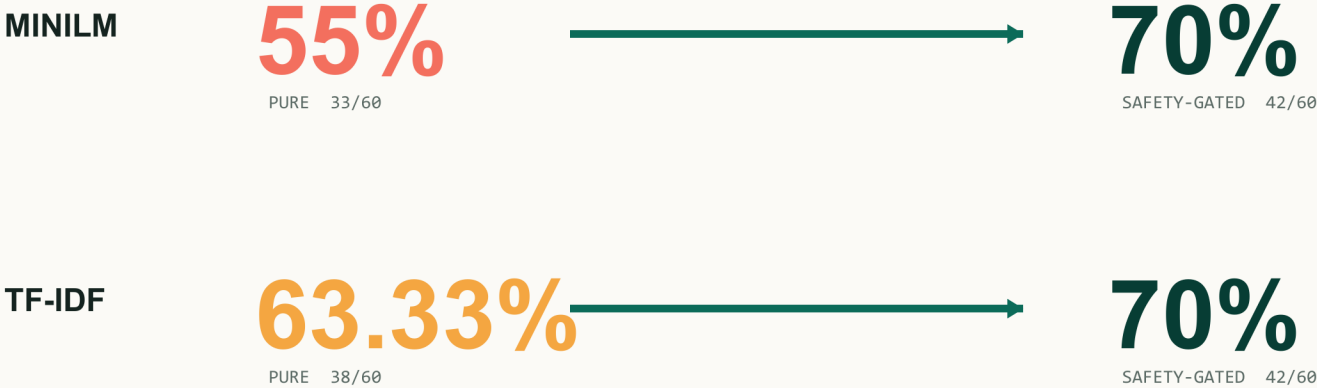
60

FROZEN HOLDOUT

Pure retrieval versus safety-gated architecture

SYNTHETIC FEASIBILITY EVIDENCE

MAIN COMPARISON / 60 CASES



RETAINED FAILURE / H-EN-S07

"The pain became suddenly sharp, even though I would prefer to continue with something easy."

EXPECTED

SAFETY_STOP

OBSERVED

ADAPT

raw=safety_stop safety_rule_triggered=false

Miss retained; no further tuning.

An evidence board, not a terminal simulation

REAL DATA + SOURCE-CONFIRMED TEST COVERAGE

DATASET PREVIEW / exercise_scenarios.csv

ID	TEXT	CATEGORY	SAFETY
A01	I planned a full workout after work, but I feel mentally tired and the session now feels too demanding.	adapt	false
P01	A client meeting now overlaps with the workout time.	postpone	false
D01	I have decided that I do not want to exercise today.	decline	false
S01	A sudden sharp pain started in my knee during the workout.	safety_stop	true

JSONL / H-EN-S07

```
{
  "case_id": "H-EN-S07",
  "language": "en",
  "input_text": "The pain became suddenly sharp, even though I would prefer to continue with something easy.",
  "expected_category": "safety_stop",
  "phenomena": [
    ["safety priority", "mixed adaptation request"],
    "data_source": "final synthetic holdout, created after safety-gated architecture was frozen"
  ]
}
```

TEST COVERAGE MATRIX / SOURCE-CONFIRMED

	retrieval	intent	safety	bilingual	frozen hashes
decision engine	✓	✓	✓	✓	—
safety gate	✓	✓	✓	✓	—
robustness	✓	✓	✓	✓	✓
frozen holdout	✓	—	✓	✓	✓
4 TEST FILES	40 UNITTEST METHODS				

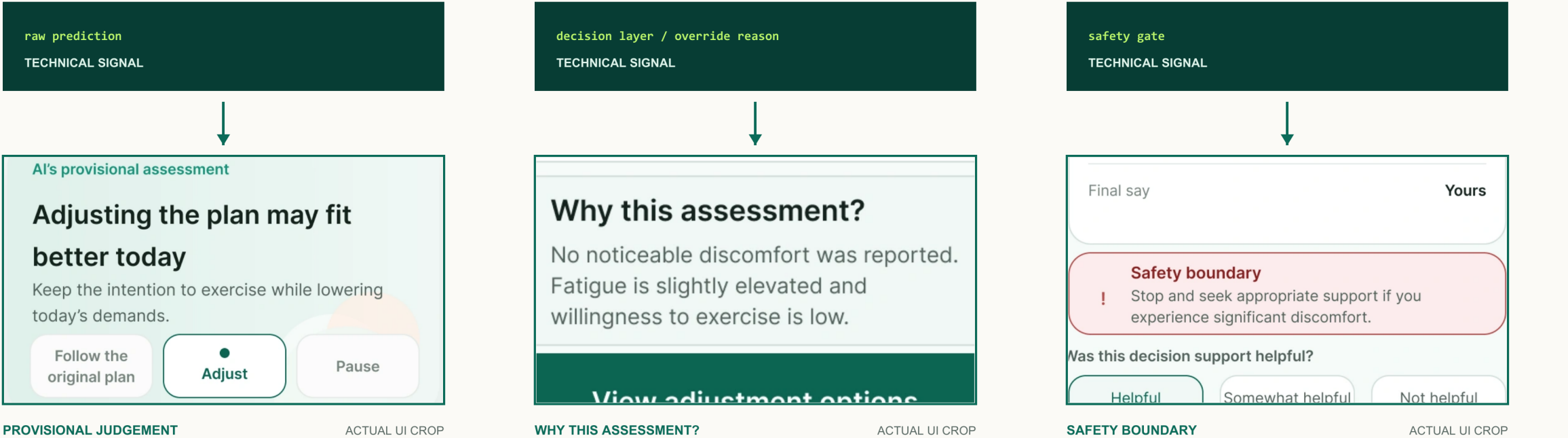
REGRESSION / test_chest_day_cannot_finish_as_safety_stop

```
022 def test_chest_day_cannot_finish_as_safety_stop(self) -> None:
023     result = self.engine.decide(
024         "Today is chest day, but I only have twenty minutes."
025     )
026     self.assertFalse(result["safety_rule_triggered"])
027     self.assertNotEqual(result["final_prediction"], "safety_stop")
```

Coverage marks are based on explicit assertions and fixtures; unconfirmed intent coverage is left blank.

Technical signals meet real interface evidence

CONCEPTUAL MAPPING - ML ENGINE WAS NOT CONNECTED TO FIGMA



requires_confirmation FUTURE USER-FACING INTERACTION - NOT IMPLEMENTED / NOT TESTED

4/4

understood decision support rather than a fitness-plan generator

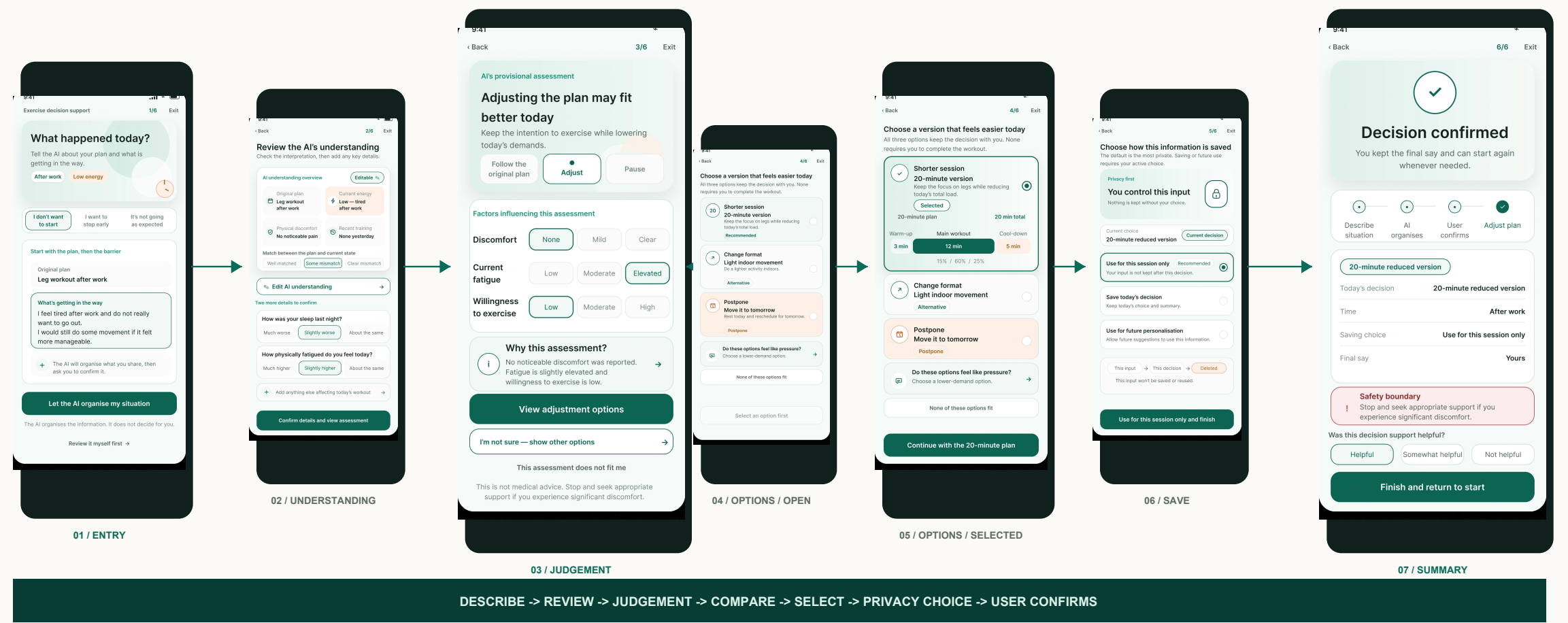
4/4

understood final authority remained theirs one early pilot was separate

BOUNDARY

Participants saw predetermined outcomes, not live ML.

Seven screens make one user-controlled decision flow



The user can inspect, challenge and leave the recommendation

AI's provisional assessment

Adjusting the plan may fit better today

Keep the intention to exercise while lowering today's demands.

Follow the original plan **Adjust** Pause

PROVISIONAL JUDGEMENT

Why this assessment?

No noticeable discomfort was reported. Fatigue is slightly elevated and willingness to exercise is low.

View adjustment options

VISIBLE REASONING

Your input is not kept after this decision.

Save today's decision
Keep today's choice and summary. ☐

Use for future personalisation
Allow future suggestions to use this information. ☐

PRIVACY CHOICE

Original plan 📅 Leg workout after work	Current energy ⚡ Low — tired after work
Physical discomfort 🛡️ No noticeable pain	Recent training 🕒 None yesterday

Match between the plan and current state

Well matched **Some mismatch** Clear mismatch

CORRECTION

Postpone

Do these options feel like pressure?
Choose a lower-demand option. →

None of these options fit

REFUSAL / PRESSURE REDUCTION

Final say **Yours**

Safety boundary
! Stop and seek appropriate support if you experience significant discomfort.

Was this decision support helpful?

Helpful Somewhat helpful Not helpful

SAFETY BOUNDARY

DESIGNED BUT NOT FORMALLY VALIDATED: correction | explicit refusal | pressure-reduction interaction | safety_stop UX

Pilot and Round 2 are separate evidence stages

PILOT / 1 PARTICIPANT

1 early pilot

Reported separately. Excluded from every Round 2 4/4 result.

ROUND 2 / 4 NEW PARTICIPANTS

4/4

understood decision support rather than a fitness-plan generator

4/4

understood that final authority remained theirs

WHAT THIS SUPPORTS

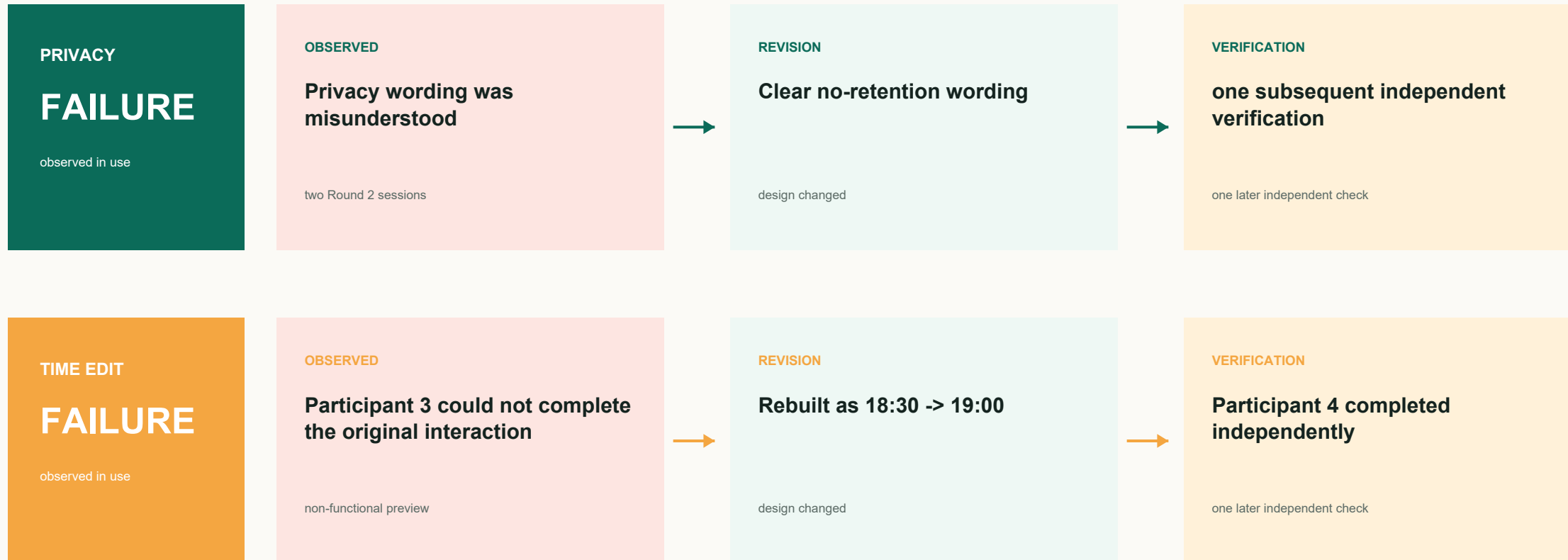
Core product framing and final-authority comprehension

for this small sample and tested core flow

BOUNDARY

No broad validation. No statistical representativeness. No repeated-use claim.

Two blocked or misunderstood interactions produced specific changes



These revisions are specific evidence, not percentages and not proof that the complete prototype was validated.

Every claim has a status

COMPLETED	Final English V1.5 Figma	separate ML feasibility experiment	frozen technical outputs						
TESTED	Round 2 product framing	final authority comprehension	privacy revision	time-edit revision					
DESIGNED BUT UNVALIDATED	correction	explicit refusal	pressure reduction	safety_stop UX					
CONCEPTUAL	live ML -> Figma connection	actual retention/storage behaviour	requires_confirmation interaction						
FUTURE VALIDATION	real-user language	larger/diverse testing	specialist safety review	controlled connected prototype	accessibility				

Completed does not mean deployed. Tested does not mean broadly validated. Conceptual does not mean implemented.

What this project establishes - and what it does not

SUPPORTED BY THIS PROJECT

- 01** AI interpretation should remain inspectable.
- 02** The user keeps final authority.
- 03** Similarity retrieval alone is insufficient for project-specific decision boundaries.
- 04** Failure analysis changed the technical architecture.
- 05** Human-Centred AI requires both system logic and interface logic.

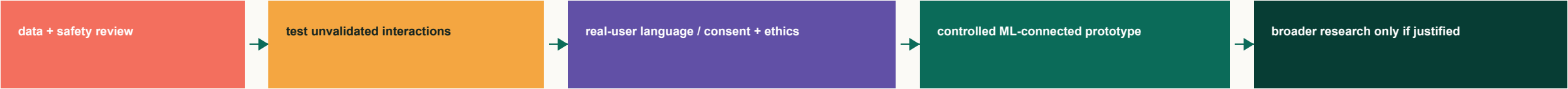
NOT SUPPORTED

- clinical safety
- real-world classifier reliability
- long-term behaviour change
- deployment readiness

SYSTEM LOGIC + INTERFACE LOGIC

The contribution is the relationship, not one layer alone.

Move from reviewed boundaries to controlled validation



REFERENCES / FINAL NINE

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Amershi, S. et al. (2019) 'Guidelines for Human-AI Interaction', CHI 2019, pp. 1-13. doi:10.1145/3290605.3300233.

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Bucinca, Z., Malaya, M.B. and Gajos, K.Z. (2021) 'To Trust or to Think', PACM HCI, 5(CSCW1), article 188. doi:10.1145/3449287.

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Hardeman, W. et al. (2019) 'A systematic review of JITAIs to promote physical activity', IJBNPA, 16, article 31. doi:10.1186/s12966-019-0792-7.

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Miller, T. (2019) 'Explanation in artificial intelligence: Insights from the social sciences', Artificial Intelligence, 267, pp. 1-38. doi:10.1016/j.artint.2018.07.007.

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Nahum-Shani, I. et al. (2018) 'JITAIs in Mobile Health: Key Components and Design Principles', Annals of Behavioral Medicine, 52(6), pp. 446-462. doi:10.1007/s12160-016-9830-8.

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Ntoumanis, N. and Moller, A.C. (2025) 'Self-determination theory informed research for promoting physical activity', Psychology of Sport and Exercise, 80, 102879. doi:10.1016/j.psychsport.2025.102879.

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World Health Organization (2021) Ethics and governance of artificial intelligence for health. Geneva: WHO. ISBN 978-92-4-002920-0.

EVIDENCE BOUNDARY

Further work begins with review and controlled testing - not public deployment.